

Background: Design

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Highlights

- An example of how mental models can make for entertainment.
- Embodied, ubiquitous, embedded, ambient, and situated interaction.
- Participatory design.

19.1 THIS IS A REFERENCE CHAPTER

This chapter contains reference material relevant to the other chapters of [Part 3](#). This chapter is not intended to be read through as a regular chapter, but each section is supposed to be read when a reference to it in the main chapters is encountered.

19.2 PARTICIPATORY DESIGN

19.2.1 Overview

Although we did not describe participatory design as a specific technique in [Chapter 14](#) (on generative design creation), we wholeheartedly support user and client participation in the entire design process, starting from ideation and sketching through to refinement. Because the specific technique of participatory design is an important part of HCI history and literature, we touch on it here.

At the very beginning of a design project, you often have the user and customers on one side and system designers on the other. Participatory design is a way to combine the knowledge of work practice of the users and customers with the process, technology, and design skills of the UX and software engineering (SE) teams.

A participatory design session usually starts with reciprocal learning in which the users and the designers learn about each other's roles; designers learn about work practices and users learn about technical constraints ([Carmel, Whitaker, &](#)

Participatory design

A democratic process for UX design actively involving all stakeholders (e.g., employees, partners, customers, citizens, users) to help ensure that the result meets their needs and is usable. Based on the argument that users should be involved in designs they will be using, and that all stakeholders, including and especially users, have equal input into UX design ([Section 11.3.4](#)).

George, 1993). The session itself is a democratic process. Rank or job title has no effect; anyone can post a new design idea or change an existing feature. Only positive and supportive attitudes are tolerated. No one can criticize or attack another person or their ideas. This leads to an atmosphere of freedom to express even the farthest out ideas; creativity rules.

In our own experience, we have found participatory design very effective for specific kinds of interaction situations. For example, we think it could be a good approach, especially if used in conjunction with design scenarios, for sketching out the first few levels of screens of the Ticket Kiosk System interaction. These first screens are very important to the user experience, where first impressions are formed by users and where we can least afford to have users get lost and customers turn away. However, in our experience, the technique sometimes does not scale up well to complete designs of large and complex systems.

For information on the history and origins of participatory design, see the next section.

For a description of PICTIVE, an example of an approach to participatory design, see the section after next.

19.2.2 History and Origins of Participatory Design

Participatory design entails user participation in design for work practice. Participatory design is a democratic process for design (social and technological) of systems involving human work, based on the argument that users should be involved in designs they will be using, and that all stakeholders, including and especially users, have equal input into interaction design (Muller & Kuhn, 1993).

The idea of user participation in system design harkens back (as does the work on contextual studies) at least to a body of effort called work activity theory (Bødker, 1991; Ehn, 1990). Originating in Russia and Germany, it flourished in Scandinavia in the 1980s where it was closely related to the workplace democracy movement. These early versions of participatory design embraced a view of design based on work practice situated in a worker's own complete environment, but also espoused empowerment of workers to "codetermine the development of the information system and of their workplace" (Clement & Besselaar, 1993).

Going back to the 1980s and earlier, probably the most well-known participatory design project was the Scandinavian project called UTOPIA (Bødker, Ehn, Kammersgaard, Kyng, & Sundblad, 1987). A main goal of Project UTOPIA was to overcome limitations on opportunities for workers to affect workplace technology and organizational work practices. UTOPIA was one of the first such projects intended to produce a commercial product at the end of the day.

Participatory design has been practiced in many different forms with different rules of engagement. In some projects, participatory design limits user power to creating only inputs for the professional designers to consider, an approach called consultative design by [Mumford \(1981\)](#). Other approaches give the users full power to share in the responsibility for the final outcome, in what Mumford calls consensus design.

Also beginning in the 1970s and 1980s, an approach to user involvement in design (but probably developed apart from the participatory design history in Scandinavia) called Joint Application Design was emerging from IBM in the United States and Canada (August 1991). Joint Application Design falls between consultative design and consensus design in the category of representative design ([Mumford, 1981](#)), a commonly used approach in industry in which user representatives become official members of the design teams, often for the duration of the project. In comparison with participatory design, Joint Application Design is often a bit more about group dynamics, brainstorming, and organized group meetings.

In the early 1990s, the Scandinavian approach to democratic design was adapted and extended within the HCI community in the form of participatory design. [Muller's \(1991\)](#) vision of participatory design as embodied in his PICTIVE approach is the most well-known adaptation of the general concept specifically to HCI. The first Participatory Design Conference met in 1990 and it has been held biannually ever since. Participatory design has since been codified for practice ([Greenbaum & Kyng, 1991](#)), reviewed ([Clement & Besselaar, 1993](#)), and summarized ([Muller, 2003](#)).

19.2.3 PICTIVE¹—An Example of an Approach to Participatory Design

Inspired by the mockup methods of the Scandinavian project called UTOPIA ([Bødker et al., 1987](#)), which provided opportunities for workers to give input to workplace technology and organizational work practices, PICTIVE ([Muller, 1991](#); [Muller, Wildman, & White, 1993b](#)) is an example of how participatory design has been operationalized in HCI. PICTIVE supports rapid group prototype design using paper and pencil and other “low technology” materials on a large tabletop in combination with video recording.

The objective is for the group to work together to find technological design solutions to support work practice and, sometimes, to redesign the work practice

¹Plastic Interface for Collaborative Technology Initiatives through Video Exploration.

in the process. Video recording is used to chronicle and communicate the design process and to record walkthroughs used to summarize the designs.

PICTIVE is, as are most participatory design approaches, a hands-on design-by-doing technique using low-tech tools, such as those used for paper prototyping: blackboards, large sheets of paper, bulletin boards, push pins, Post-it notes, colored marking pens, index cards, scissors, and tape. PICTIVE deliberately uses these low-tech (noncomputer, nonsoftware) representations to level the playing field between users and technical design team members. Otherwise, using even the most primitive programming tools for building prototypes on the fly can cast the users as outsiders and the design practitioners as intermediaries through whom all user ideas must flow. It then is no longer a collaborative storytelling activity.

After the mutual introduction to each others' backgrounds and perspectives, the group typically discusses the task at hand and the design objectives to get on the same page for doing the design. Then they gather around a table on which there is a large paper representation of a generic computer "window." Anyone can step forward and "post" a design feature, for example, button, icon, menu, dialogue box, or message, by writing or drawing it on a Post-it note or similar piece of paper, sticking it on the "window" working space, and explaining the rationale. The group can then discuss refinements and improvements. Someone else can edit the text on the object, for example, and change its location in the window.

The group works collaboratively to expand and modify, adding new objects, changing objects, and moving objects to create new layouts and groupings and changing wording of labels and messages, and so on, all the while communicating their thinking and reasons behind each change. The results can be evaluated immediately as low-fidelity prototypes with walkthroughs (usually recorded as video for further sharing and evaluation). In most project environments that use this kind of participatory design, it is often used in the consultative design mode, where users participate in forming parts of the design but the professional design practitioners have the final responsibility for the overall design.

PICTIVE has been evaluated informally in the context of several real product design projects (Muller, 1992). User participants report getting enjoyment from the process and great satisfaction in having a receptive audience for their own design ideas and, especially, in seeing those design ideas included in the group's output.

19.3 MENTAL MODELS: AN EXAMPLE OF HOW THEY CAN MAKE FOR ENTERTAINMENT

Lack of a correct user mental model can be the stuff of comedy curve balls. An example is the scene in the 1992 movie, *My Cousin Vinny*, where Marisa Tomei—as Vinny’s fiancée, Mona Lisa Vito—tries to make a simple phone call. This fish-out-of-water scene pits a brash young woman from New York against a rotary dial telephone in a backwater Southern town. You cannot help but reflect on the mismatch in the mapping between her mental model of touch-tone operation and the reality of old-fashioned rotary dials as she pokes vigorously at the numbers through the finger holes.

But, lest you dismiss her as completely out of touch, we remind you that it was she who solved the case with her esoteric automobile knowledge, proving that the suspects’ 1964 Buick Skylark could not have left the two tire tracks found outside the convenience store because it did not have a limited-slip differential.